

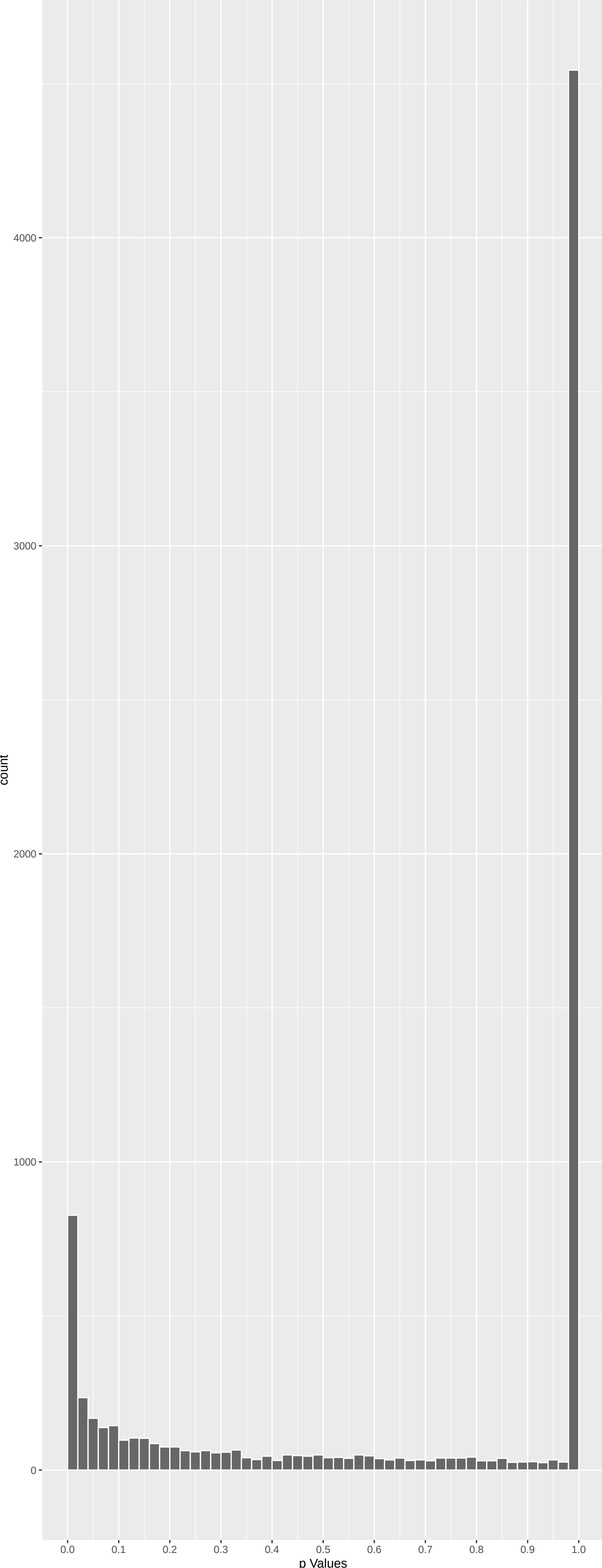
Top genes by P-value non-permulated

| Gene     | Rho       | P            | p.adj     | qValueNoperm |
|----------|-----------|--------------|-----------|--------------|
| CPB1     | 7.869619  | 2.134329e-14 | 3.224e-10 | 4.870e-06    |
| ADGRF1   | 7.124748  | 6.256290e-12 | 4.725e-08 | 3.569e-04    |
| CLDN16   | 6.861753  | 4.081239e-11 | 1.244e-07 | 3.758e-04    |
| TRPM6    | 6.860483  | 4.117700e-11 | 1.244e-07 | 3.758e-04    |
| SLC5A12  | 6.865191  | 3.984132e-11 | 1.244e-07 | 3.758e-04    |
| PGC      | 6.669847  | 1.536419e-10 | 3.868e-07 | 9.738e-04    |
| CPLX3    | -6.557372 | 3.285867e-10 | 7.090e-07 | 1.530e-03    |
| CFAP20DC | 6.478637  | 5.553281e-10 | 1.049e-06 | 1.946e-03    |
| CRYAA    | -6.445609 | 6.908229e-10 | 1.159e-06 | 1.946e-03    |
| SLC14A2  | 6.137220  | 5.038698e-09 | 7.611e-06 | 1.150e-02    |
| SPG11    | 6.069822  | 7.683109e-09 | 1.055e-05 | 1.449e-02    |
| CTRC     | 6.035824  | 9.489220e-09 | 1.194e-05 | 1.504e-02    |
| CRYAA2   | -6.007193 | 1.132580e-08 | 1.316e-05 | 1.525e-02    |
| CCDC66   | 5.983563  | 1.309855e-08 | 1.413e-05 | 1.525e-02    |
| DST      | 5.912238  | 2.024943e-08 | 2.039e-05 | 2.005e-02    |
| MUC20    | 5.857279  | 2.823082e-08 | 2.389e-05 | 2.005e-02    |
| ACADM    | 5.857847  | 2.813432e-08 | 2.389e-05 | 2.005e-02    |
| JAKMIP1  | -5.855889 | 2.846790e-08 | 2.389e-05 | 2.005e-02    |
| PRRI4L   | 5.831189  | 3.302036e-08 | 2.625e-05 | 2.040e-02    |
| BPIFB1   | 5.817875  | 3.576029e-08 | 2.701e-05 | 2.040e-02    |
| DHR54    | 5.777860  | 4.539398e-08 | 3.157e-05 | 2.168e-02    |
| ACY1     | 5.775696  | 4.598415e-08 | 3.157e-05 | 2.168e-02    |
| SLC22A13 | 5.725911  | 6.172814e-08 | 4.054e-05 | 2.662e-02    |
| TMEM131L | 5.707478  | 6.879735e-08 | 4.330e-05 | 2.696e-02    |
| TMED6    | 5.695396  | 7.385135e-08 | 4.462e-05 | 2.696e-02    |

Top genes by Q-Value non-permulated

| Gene     | Rho       | P            | p.adj     | qValueNoperm |
|----------|-----------|--------------|-----------|--------------|
| CPB1     | 7.869619  | 2.134329e-14 | 3.224e-10 | 4.870e-06    |
| ADGRF1   | 7.124748  | 6.256290e-12 | 4.725e-08 | 3.569e-04    |
| CLDN16   | 6.861753  | 4.081239e-11 | 1.244e-07 | 3.758e-04    |
| TRPM6    | 6.860483  | 4.117700e-11 | 1.244e-07 | 3.758e-04    |
| SLC5A12  | 6.865191  | 3.984132e-11 | 1.244e-07 | 3.758e-04    |
| PGC      | 6.669847  | 1.536419e-10 | 3.868e-07 | 9.738e-04    |
| CPLX3    | -6.557372 | 3.285867e-10 | 7.090e-07 | 1.530e-03    |
| CRYAA    | -6.445609 | 6.908229e-10 | 1.159e-06 | 1.946e-03    |
| CFAP20DC | 6.478637  | 5.553281e-10 | 1.049e-06 | 1.946e-03    |
| SLC14A2  | 6.137220  | 5.038698e-09 | 7.611e-06 | 1.150e-02    |
| SPG11    | 6.069822  | 7.683109e-09 | 1.055e-05 | 1.449e-02    |
| CTRC     | 6.035824  | 9.489220e-09 | 1.194e-05 | 1.504e-02    |
| CCDC66   | 5.983563  | 1.309855e-08 | 1.413e-05 | 1.525e-02    |
| CRYAA2   | -6.007193 | 1.132580e-08 | 1.316e-05 | 1.525e-02    |
| MUC20    | 5.857279  | 2.823082e-08 | 2.389e-05 | 2.005e-02    |
| DST      | 5.912238  | 2.024943e-08 | 2.039e-05 | 2.005e-02    |
| ACADM    | 5.857847  | 2.813432e-08 | 2.389e-05 | 2.005e-02    |
| JAKMIP1  | -5.855889 | 2.846790e-08 | 2.389e-05 | 2.005e-02    |
| PRRI4L   | 5.831189  | 3.302036e-08 | 2.701e-05 | 2.040e-02    |
| BPIFB1   | 5.817875  | 3.576029e-08 | 2.701e-05 | 2.040e-02    |
| DHR54    | 5.777860  | 4.539398e-08 | 3.157e-05 | 2.168e-02    |
| ACY1     | 5.775696  | 4.598415e-08 | 3.157e-05 | 2.168e-02    |
| SLC22A13 | 5.725911  | 6.172814e-08 | 4.054e-05 | 2.662e-02    |
| TMED6    | 5.695396  | 7.385135e-08 | 4.462e-05 | 2.696e-02    |
| TMEM131L | 5.707478  | 6.879735e-08 | 4.330e-05 | 2.696e-02    |

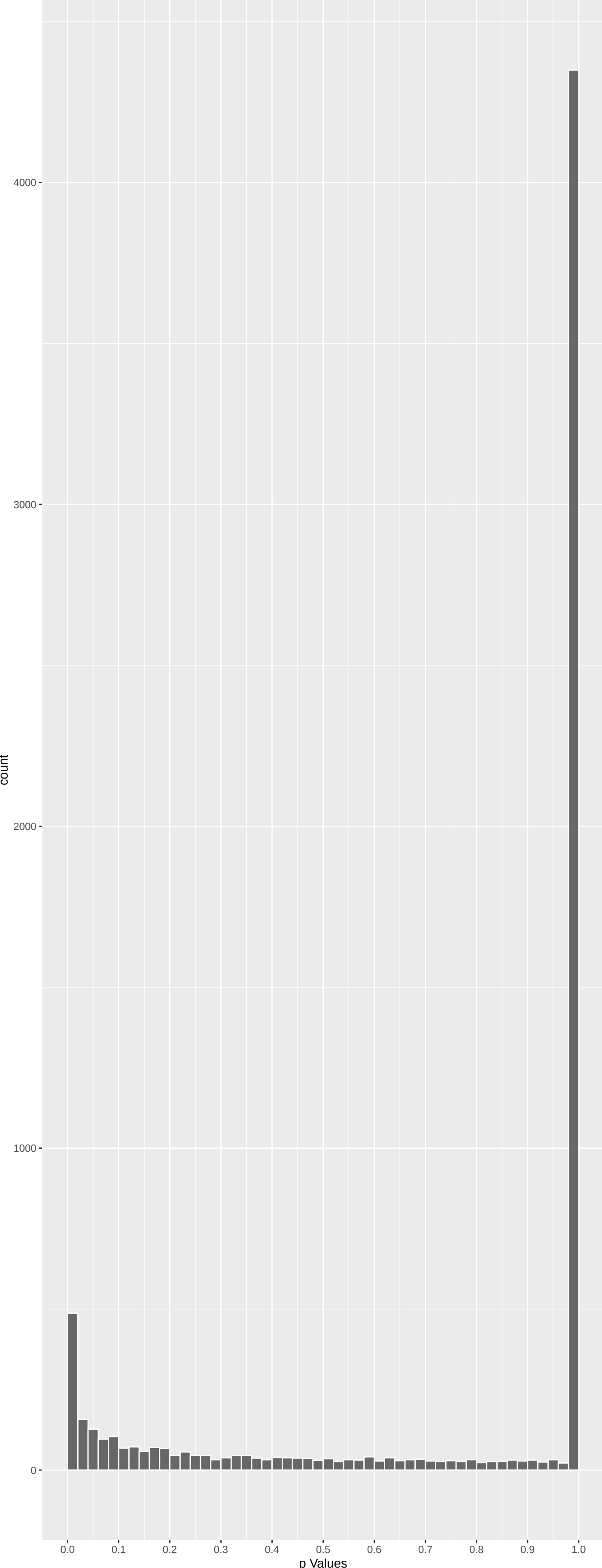
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

| Gene     | Rho      | P            | p.adj     | qValueNoperm |
|----------|----------|--------------|-----------|--------------|
| CPB1     | 7.869619 | 2.134329e-14 | 3.224e-10 | 4.870e-06    |
| ADGRF1   | 7.124748 | 6.256290e-12 | 4.725e-08 | 3.569e-04    |
| CLDN16   | 6.861753 | 4.081239e-11 | 1.244e-07 | 3.758e-04    |
| TRPM6    | 6.860483 | 4.117700e-11 | 1.244e-07 | 3.758e-04    |
| SLC5A12  | 6.865191 | 3.984132e-11 | 1.244e-07 | 3.758e-04    |
| PGC      | 6.669847 | 1.536419e-10 | 3.868e-07 | 9.738e-04    |
| CFAP20DC | 6.478637 | 5.553281e-10 | 1.049e-06 | 1.946e-03    |
| SLC14A2  | 6.137220 | 5.038698e-09 | 7.611e-06 | 1.150e-02    |
| SPG11    | 6.069822 | 7.683109e-09 | 1.055e-05 | 1.449e-02    |
| CTRC     | 6.035824 | 9.489220e-09 | 1.194e-05 | 1.504e-02    |

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

| Gene    | Rho       | P            | p.adj     | qValueNoperm |
|---------|-----------|--------------|-----------|--------------|
| CPLX3   | -6.557372 | 3.285867e-10 | 7.090e-07 | 1.530e-03    |
| CRYAA   | -6.445609 | 6.908229e-10 | 1.159e-06 | 1.946e-03    |
| CRYAA2  | -6.007193 | 1.132580e-08 | 1.316e-05 | 1.525e-02    |
| JAKMIP1 | -5.855889 | 2.846790e-08 | 2.389e-05 | 2.005e-02    |
| PRKCG   | -5.647347 | 9.776547e-08 | 5.487e-05 | 2.976e-02    |
| EP400   | -5.578507 | 1.455548e-07 | 6.679e-05 | 2.976e-02    |
| TRPM1   | -5.562257 | 1.597849e-07 | 6.896e-05 | 2.976e-02    |
| CRYBB2  | -5.538820 | 1.827102e-07 | 7.666e-05 | 3.217e-02    |
| CAMK2B  | -5.482036 | 2.522757e-07 | 9.294e-05 | 3.424e-02    |
| H4-16   | -5.484907 | 2.482118e-07 | 9.294e-05 | 3.424e-02    |



GO\_Biological\_Process\_2023 Top pathways by non-permutation

| Geneset                                    | stat        | num.genes | pval      | p.adj     | gene.vals                                                                           |
|--------------------------------------------|-------------|-----------|-----------|-----------|-------------------------------------------------------------------------------------|
| tRNA Methylation (GO:0030408)              | 0.25364110  | 34        | 3.150e-07 | 1.692e-03 | TARBP1:75 TRMT9B:459 TRMT1L:460 LCM2:744 TRMT10A:503 TRMT5:538 TRMT1:643            |
| Fatty Acid Catabolic Process (GO:0009062)  | 0.18065022  | 55        | 3.695e-06 | 9.922e-03 | EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 EC2:180 HADHA:245 CP2T:314                   |
| Chemical Synaptic Transmission (GO:00072)  | -0.09386238 | 188       | 1.008e-05 | 1.353e-02 | GABRR1:16 STXBP1:47 GABRB3:122 RMRP2:128 RPS6K2:138 CHRFM7A:139 PTEN:151            |
| Regulation Of Synaptic Transmission, Glu   | -0.20396950 | 40        | 8.215e-06 | 1.353e-02 | DGKI:24 PTK2B:63 MAPKBIP2:109 SHANK1:342 CACNG3:159 SYT1:199 GRM5:467               |
| Alpha-Amino Acid Catabolic Process (GO:1)  | 0.22897878  | 29        | 1.998e-05 | 1.789e-02 | ALDH4A1:80 TAT:187 HIBCH:219 KMO:432 COT2:358 ACSMD:591 KYNU:973                    |
| Mitochondrial Gene Expression (GO:014005)  | 0.13059664  | 91        | 1.743e-05 | 1.789e-02 | PTCD3:67 TEFM:95 MRPS9:243 MRPL4:345 MRPS18A:504 MRPS22:599 FTB2M:604               |
| NADH Dehydrogenase Complex Assembly (GO:1) | 0.18870308  | 41        | 2.957e-05 | 1.925e-02 | EC5IT:79 FOXRED1:98 NDUF9A:169 NDUF8B:308 DMAC2:347 TMEM1268:499 NDUF86:532         |
| RNA Methylation (GO:0001510)               | 0.17131883  | 48        | 4.105e-05 | 1.925e-02 | TARBP1:75 TRMT9B:459 LCM2:744 RBM15B:480 TRMT10A:503 FTB2M:604 NSUN6:657            |
| Fatty Acid Beta-Oxidation (GO:0006635)     | 0.17779355  | 45        | 3.757e-05 | 1.925e-02 | EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 AC0X2:174 EC12:180 HADHA:245                 |
| Mitochondrial Respiratory Chain Complex    | 0.18870308  | 41        | 2.957e-05 | 1.925e-02 | EC5IT:79 FOXRED1:98 NDUF9A:169 NDUF8B:308 DMAC2:347 TMEM1268:499 NDUF86:532         |
| Non-Motile Cilium Assembly (GO:1905515)    | 0.22186584  | 29        | 3.587e-05 | 1.925e-02 | CEP89:210 ARL13B:282 CYP250:666 GORAB:887 ENMCI:903 CC2D3:1040 MBD1:1264            |
| Steroid Metabolic Process (GO:0008202)     | 0.15825673  | 56        | 4.301e-05 | 1.925e-02 | CYP2R1:21 CYP1A1:72 CYP2B6:138 DHRS4:194 CYP17A1:225 NPC1:277 CYP11A1:443           |
| Organic Hydroxy Compound Transport (GO:0)  | 0.19316430  | 36        | 6.141e-05 | 2.537e-02 | ABCG5:141 SLC51A:211 SLC51B:334 RALBP1:404 STRA6:466 SLC01A2:549 SLC10A2:873        |
| Cilium Organization (GO:0044782)           | 0.08572650  | 183       | 6.897e-05 | 2.646e-02 | TCTN1:34 CDC66:71 CFAP43:116 TCTN2:182 CEP89:210 IQUB:251 NPHP3:259                 |
| Mitochondrial Translation (GO:0032543)     | 0.12107935  | 87        | 9.829e-05 | 3.519e-02 | PTCD3:67 MRPS9:243 MTIF2:269 MRPL4:345 GFM1:452 MRPS18A:504 MRPS22:599              |
| Monocarboxylic Acid Transport (GO:001571)  | 0.15177899  | 50        | 1.157e-04 | 3.656e-02 | SLC5A12:5 SLC16A5:183 SLC22A13:205 SLC51A:211 SLC51B:334 SLC61A:241 SLC16A9:420     |
| tRNA Modification (GO:0006400)             | 0.14344262  | 61        | 1.094e-04 | 3.656e-02 | TARBP1:75 TRMT9B:459 LCM2:744 TRMT10A:503 TRMT1:643 NSUN6:657 MTO1:746              |
| Mitochondrial Respiratory Chain Complex    | 0.19323841  | 62        | 1.532e-04 | 4.167e-02 | EC5IT:79 FOXRED1:98 NDUF9A:169 NDUF8B:308 DMAC2:347 TMEM1268:499 NDUF86:532         |
| Modulation Of Chemical Synaptic Transmis   | -0.15636539 | 89        | 1.480e-04 | 4.167e-02 | DGKI:24 TRIO:77 MAPKBIP2:109 PTEN:151 GRM1:172 BA1P3:176 SYTL:199                   |
| Positive Regulation Of Synaptic Transmis   | -0.15330835 | 51        | 1.552e-04 | 4.167e-02 | PTK2B:63 SHANK1:34 CACNG3:159 BA1P3:176 CHRHR2:186 SHANK2:188 SYTL:199              |
| Protein Glycosylation (GO:0006486)         | 0.10449613  | 107       | 1.961e-04 | 5.016e-02 | TET1:7 B3GNTA:247 GALNT5:205 POGTLU1:257 NPC1:277 FKTN:291 TZ3GAL5:325              |
| Double-Strand Break Repair (GO:0006302)    | 0.09222659  | 134       | 2.397e-04 | 5.852e-02 | ANKL1:168 WRN:207 ERCC5:259 NPC1:277 RNKFB6:286 PRKDC:671 RNF169:612                |
| Cell Junction Assembly (GO:0034329)        | -0.13319888 | 63        | 2.617e-04 | 6.112e-02 | SDK1:80 TLN1:94 SHANK1:346 PTEN:151 SHANK2:188 DPNL:295 CDH6:328                    |
| Cilium Assembly (GO:0060271)               | 0.07535552  | 197       | 2.867e-04 | 6.333e-02 | TCTN1:34 CDC66:71 CFAP43:116 TCTN2:182 CDC39:197 CEP89:210 IQUB:251                 |
| Nervous System Development (GO:0007399)    | -0.06116961 | 301       | 2.948e-04 | 6.333e-02 | VAX2:8 ADGRB1:27 SH3GL3:43 SCIN:60 CNTN4:70 SDK1:80 HDAC4:118                       |
| Monocarboxylic Acid Catabolic Process (G   | 0.24581269  | 18        | 3.071e-04 | 6.344e-02 | CYP26C1:38 AGXT:626 AGXT2:1549 HOGA1:1721 IDUA:1865 LPN3:1890 NUDT19:2051           |
| Muscle Contraction (GO:0006936)            | -0.05465004 | 72        | 3.819e-04 | 7.598e-02 | MYH13:5 TACR2:20 MYH2:116 LMOD1:267 ACTA2:74 PTPRE:61 PTK2B:63 TMED2:71 SRPRA:84    |
| Intergrade Trans-Synaptic Signaling (GO)   | -0.08793602 | 134       | 4.616e-04 | 8.717e-02 | GABRR1:16 GABRB3:122 RPS6K2:138 SYTL:199 KCNQ3:245 RNF483:326 CHRNA4:336            |
| Signal Release From Synapse (GO:0099643)   | -0.18778908 | 29        | 4.707e-04 | 8.717e-02 | STXBP1:47 SYTL:199 SYC32A1:218 HRH3:326 CPLX1:346 NRXN:524 XNP2:514                 |
| Fatty Acid Oxidation (GO:0019395)          | 0.15246630  | 43        | 5.489e-04 | 9.828e-02 | EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 EC2:180 HADHA:245 CP2T:314                   |
| Bile Acid And Bile Salt Transport (GO:00)  | 0.24608941  | 16        | 5.698e-04 | 1.080e-01 | SLC51A:211 SLC51B:334 SLC01A2:549 SLC10A2:873 ATP8B1:1063 SLC01B1:1345 SLC10A5:1649 |
| Central Nervous System Development (GO:0)  | -0.06807124 | 213       | 6.638e-04 | 1.080e-01 | VAX2:8 SH3GL3:43 SCIN:60 CNTN4:70 PTEN:151 SHROOM2:168 DCX:348                      |
| Double-Strand Break Repair Via Homologou   | 0.10413484  | 90        | 6.581e-04 | 1.080e-01 | FANCM:81 WRN:207 ERCC5:259 NPC1:277 RNF01:612 RNF169:612 RAD54B:807 RPSV:876        |
| Glutamate Receptor Signaling Pathway (GO)  | -0.19194577 | 26        | 7.097e-04 | 1.100e-01 | PTK2B:63 HOMER2:325 TRPM1:356 GRM5:467 GRM5:636 GRIN2A:593 GRM8:693                 |
| Recombinational Repair (GO:0007025)        | 0.11246635  | 76        | 7.165e-04 | 1.100e-01 | WRN:207 ERCC5:259 NPC1:277 RNF01:612 RNF169:612 RAD54B:807 RPSV:876                 |
| Fatty Acid Metabolic Process (GO:0006631)  | 0.09606542  | 103       | 7.817e-04 | 1.166e-01 | ABCB11:65 CYP1A1:72 ACADM:92 CPT1C:97 PLA2G3A:166 AC0X2:174 ACADSB:293              |
| Negative Regulation Of Fatty Acid Biosyn   | 0.36494792  | 7         | 8.274e-04 | 1.201e-01 | TRIB3:315 ACADVL:362 BRCA1:932 UBR4:2343 DCAF5:274 SIRT1:2866 PIBF1:2909            |
| Fat-Soluble Vitamin Metabolic Process (G   | 0.23922218  | 6         | 9.266e-04 | 1.214e-01 | CYP2R1:21 CYP1A1:72 CYP11A1:443 GGCK:571 CYP24A1:655 LATR:802 PIAS4:972             |
| Nitrogen Compound Transport (GO:0071705)   | 0.08351818  | 133       | 9.200e-04 | 1.214e-01 | CUBN:20 RHAG:185 SLC22A13:205 AP09:278 RHC2:31 SLC6A:256 RH0:313                    |
| Positive Regulation Of Synaptic Transmis   | -0.26609819 | 13        | 8.963e-04 | 1.214e-01 | PTK2B:63 SHANK3:146 CACNG3:159 GRIN2A:593 NPS:626 TSHZ2:724 CCR2:5124.5             |

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

| Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                                |
|------------------------------------------|-------------|-----------|-----------|-----------|--------------------------------------------------------------------------|
| REACTOME_DEVELOPMENTAL_BIOLOGY           | -0.90971584 | 680       | 1.627e-15 | 1.053e-11 | COL6A2:22 KRT20:37 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78 TLN1:94           |
| REACTOME_NERVOUS_SYSTEM_DEVELOPMENT      | -0.09599440 | 390       | 1.027e-10 | 1.326e-07 | COL6A2:22 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78 TLN1:94 SEMA5A:133         |
| REACTOME_INFECTION_DISEASE               | -0.07237993 | 556       | 7.552e-09 | 1.630e-05 | ADCY1:3 GOLGA7:68 DOCK1:38 SH3GL3:43 DYNC1H1:53 ITPR1:54 AP2A2:78        |
| HOUNKPE_HOUSEKEEPING_GENES               | -0.06326255 | 712       | 1.368e-08 | 1.772e-05 | PHRF1:32 NSA2:36 HCF1:66 SNDI1:13 U2AF1:17 PTEN:151 ABCE1:160            |
| REACTOME_NEURONAL_SYSTEM                 | -0.10094367 | 270       | 1.325e-08 | 1.772e-05 | ADCY1:3 CAMK2B:6 GABRR1:16 STXBP1:47 NBEA:51 AP2A2:78 SLITRK4:95         |
| REACTOME_CELLULAR_RESPONSES_TO_STIMULI   | -0.07356497 | 469       | 2.924e-08 | 3.156e-05 | CAMK2B:6 EP007: DNUAJA4:18 RPTOR:25 DYNC1H1:53 SRPRA:84 TLN1:94          |
| REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_ | -0.08094527 | 378       | 7.967e-08 | 6.585e-05 | COL6A2:22 DOCK1:38 SH3GL3:43 COL6A1:45 ITPR1:54 AP2A2:78 TEC:64          |
| KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER | -0.07327046 | 464       | 8.135e-08 | 6.585e-05 | CAMK2B:6 DOCK4:35 TCF15:42 STXBP1:47 DYNC1H1:53 TMED2:71 USP11:91        |
| REACTOME_SIGNALING_BY_WNT                | -0.11686042 | 161       | 3.344e-07 | 2.349e-04 | ITPR1:54 PDE6B:74 AP2A2:78 PSMA3:205 TNRC6A:221 AGO3:280 XIAP:286        |
| WP_MRNA_PROCESSING                       | -0.18062307 | 66        | 3.989e-07 | 2.349e-04 | U2AF2:117 SNRNP:156 SNRPA:161 PTBP2:163 SFP2:270 SFSWAP:301 SRSF10:354   |
| BLALOCK_ALZHEIMERS_DISEASE_DN            | -0.05465004 | 766       | 3.869e-07 | 2.349e-04 | STXBP1:47 DYNC1H1:53 ITPR1:54 PTPRE:61 PTK2B:63 TMED2:71 SRPRA:84        |
| REACTOME_PROCESSING_OF_CAPPED_INTRON_CON | -0.11198787 | 170       | 5.083e-07 | 2.743e-04 | SLBP:73 U2AF2:117 SNRNP:156 SNRPA:161 SFP2:270 SYMPK:283 RAE1:312        |
| BENPORATH_ES_WITH_H3K27ME3               | -0.05465478 | 733       | 6.619e-07 | 3.297e-04 | CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 SMPD3:23 DGKI:24 HSF4:30             |
| REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION | -0.08108675 | 297       | 7.134e-06 | 8.023e-04 | CAMK2B:6 TLN1:94 SNDI1:13 HDAC4:118 PTEN:151 PSMA3:205 MYC:324           |
| REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_ | -0.12668163 | 118       | 1.092e-06 | 9.031e-04 | ADCY1:3 CAMK2B:6 GABRR1:16 NBEA:51 AP2A2:78 GABRB3:122 RPS6K2:138        |
| CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL | 0.05013951  | 785       | 2.477e-06 | 1.003e-03 | PTGFR:6 ACY1:35 IDO2:51 ABCB11:65 CYP1A1:72 HMCE5:77 EHADH:78            |
| NIKOLSKY_BREAST_CANCER_12Q24_AMPlicON    | -0.36237930 | 14        | 2.677e-06 | 1.020e-03 | EP400:7 POLE:102 MMP1:237 GALNT9:261 SFSWAP:301 ANKLE2:394 DD51:524      |
| NIKOLSKY_BREAST_CANCER_8P12_P11_AMPlicON | 0.21621579  | 38        | 4.029e-06 | 1.450e-03 | ANK1:36 IDO2:51 RAB11FIP1:53 NSD3:64 DHDH2:111 IKK8:135 ADAM9:213        |
| REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC | 0.25453672  | 26        | 7.026e-06 | 2.275e-03 | CYP2R1:21 ACY1:35 CYP2C1:38 CYP17A1:225 CYP11A1:443 CYP24A1:655 FDXR:660 |
| REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION | -0.05052726 | 699       | 6.945e-06 | 2.275e-03 | CAMK2B:6 IGFBP3:13 RPTOR:25 SLBP:73 ZNF704:86 USP28:90 U2AF2:117         |
| WP_Q3Q7_COPY_NUMBER_VARIATION_SYNDROME   | -0.13948118 | 86        | 8.004e-06 | 2.468e-03 | KLHL30:21 KIF1A:34 GYS1:68 HDAC4:118 TRAF3P1:131 GPC1:177 PDCD1:268      |
| KIM_ALL_DISORDERS_ISALB1_CORR_UP         | -0.07010637 | 337       | 1.097e-05 | 3.088e-03 | CAMK2B:6 DOCK4:35 STXBP1:47 DYNC1H1:53 ITPR1:54 PTK2B:63 TMED2:71        |
| KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA | 0.19407544  | 43        | 1.079e-05 | 3.088e-03 | EHHADH:78 ACA2A:83 ACADM:92 ACADH:189 HIBCH:219 HADHA:245 ACADSB:293     |
| REACTOME_FATTY_ACID_METABOLISM           | -0.10783639 | 139       | 1.191e-05 | 3.213e-03 | CYP1A1:72 EHADH:78 ACA2A:83 ACADM:92 ACAD11:106 ACOT12:109 AC0X2:174     |
| NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPlicO | -0.12651016 | 100       | 1.275e-05 | 3.304e-03 | CABLES2:31 TCF15:42 ZBTB46:201 RAE1:312 HRH3:326 SYTL1:327 CHRNA4:336    |
| LIM_MAMMARY_STEM_CELL_UP                 | -0.0686132  | 364       | 1.340e-05 | 3.311e-03 | ADCY1:10 UPP1:12 IGFBP3:13 PTPRE:61 TRM5:29 GPM1:85 BNC1:96              |
| REACTOME_SIGNALING_BY_GPCR               | -0.05637608 | 517       | 1.380e-05 | 3.311e-03 | SORCS1:1 CAMK2B:6 TACR2:20 DGKI:24 GRM2:29 GAL:40 NBEA:51                |
| MIKKELSEN_MEF_HCP_WITH_H3K27ME3          | -0.06080085 | 435       | 1.555e-05 | 3.597e-03 | CAMK2B:6 ADGRB1:27 GAL:40 CNTN4:70 MAPKBIP2:109 OTUD7A:114 GRID1:172     |
| BENPORATH_SUZ12_TARGETS                  | -0.04873901 | 684       | 1.774e-05 | 3.844e-03 | VAX2:8 SORCS1:10 DGKI:24 HSF4:30 RPS2:41 RNF138:287 IGA:75               |
| BENPORATH_EED_TARGETS                    | -0.04902705 | 675       | 1.781e-05 | 3.844e-03 | CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 DGKI:24 HSF4:30 SORCS2:33            |
| REACTOME_INFLUENZA_INFECTION             | -0.13931432 | 79        | 1.906e-05 | 3.982e-03 | RAE1:312 RPL7A:408 RPS48:445 NUP93:645 NUP37:518 KPNAB:564 ISG15:607     |
| REACTOME_BIOLOGICAL_OXIDATIONS           | -0.11625754 | 123       | 1.981e-05 | 4.008e-03 | CYP2R1:21 ACY1:35 ALDH1A1:60 CYP1A1:72 CYP2C6:138 MGST3:215 PODXL:226    |
| REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY | -0.09439264 | 170       | 2.296e-05 | 4.506e-03 | ADCY1:3 CAMK2B:6 GABRR1:16 STXBP1:47 NBEA:51 AP2A2:78 GABRB3:122         |
| WP_MALIGNANT_PLEURAL_MESOTHELIOMA        | -0.07047266 | 301       | 2.856e-05 | 5.465e-03 | RPTOR:25 TRAF2:55 HCF1:66 PTEN:151 PTK1:71 FZD10:198 ULK2:201            |
| LEIN_NEURON_MARKERS                      | -0.10704696 | 47        | 5.554e-05 | 1.028e-02 | JAKMIP1:1 CAMK2B:6 RPS2:41 SNRNP:156 SYTL:199 DPP6:200 TRANK1:274        |
| REACTOME_HIV_INFECTION                   | -0.09758035 | 142       | 6.206e-05 | 1.032e-02 | AP2A2:78 PSMA3:205 RAE1:312 NUP93:645 FURIN:516 NUP37:518 ELMO1:555      |
| REACTOME_GENE_SILENCING_BY_RNA           | -0.13944262 | 69        | 6.304e-05 | 1.032e-02 | MOV10L1:115 TNRC6A:221 AGO3:280 RAE1:312 NUP93:645 NUP37:518 H2BC15:678  |
| REACTOME_TRANSPORT_OF_MATUR_TRANSCRIPT_  | -0.15351799 | 57        | 6.185e-05 | 1.032e-02 | SLBP:73 U2AF2:117 SYMPK:283 RAE1:312 SLU7:380 NUP93:645 NUP37:518        |
| WONG_MITOCHONDRIA_GENE_MODULE            | 0.09467620  | 151       | 6.191e-05 | 1.032e-02 | CHDH:32 CRVZ:70 NDUF9A:169 SLC16A5:183 ATP13A3:249 GCDH:286 ACADSB:293   |
| KEGG_SPLICEOSOME                         | -0.15742537 | 54        | 6.374e-05 | 1.032e-02 | U2AF2:117 SNRPA:161 SFP3A:270 SMNDP1:331 SRSF10:354 SLU7:380 HSP8A:409   |

DisGeNET Top pathways by non-permutation

| Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                                      |
|------------------------------------------|-------------|-----------|-----------|-----------|--------------------------------------------------------------------------------|
| Mitochondrial Diseases                   | 0.07320435  | 308       | 1.173e-05 | 1.072e-01 | ALB:19 TXNRD1:48 CRVZ:70 EC5IT:79 FOXRED1:98 NDUF9A:169 CEP89:210              |
| Schizophrenia                            | -0.03664206 | 1278      | 2.194e-05 | 1.072e-01 | ADCY1:3 CAMK2B:6 GABRR1:16 DOCK4:35 TMEM132D:39 STXBP1:47 CNTN4:70             |
| Central neuroblastoma                    | -0.03773073 | 1086      | 4.686e-05 | 1.140e-01 | CAMK2B:6 IGFBP3:13 KIF1A:34 DOCK4:35 GAL:40 ITPR1:54 CRABP1:57                 |
| Neuroblastoma                            | -0.03760430 | 1113      | 4.095e-05 | 1.140e-01 | CAMK2B:6 IGFBP3:13 KIF1A:34 DOCK4:35 GAL:40 ITPR1:54 CRABP1:57                 |
| HIV Infections                           | -0.05230156 | 515       | 6.165e-05 | 1.204e-01 | KRT20:37 TMEM132D:39 PTK2B:63 TMED2:71 VIPR1:103 PLEC:142 ABCE1:160            |
| Cholestasis                              | 0.07462788  | 229       | 1.098e-04 | 1.532e-01 | ABCB11:65 SLC23A1:68 ABCG5:141 PLA2G1B:146 SLC51A:211 CYP17A1:225 HADHA:245    |
| Neurodevelopmental Disorders             | -0.09339233 | 146       | 1.041e-04 | 1.532e-01 | STXBP1:47 ITPR1:54 HCF1:66 CNTN4:70 TRIO:77 GABRB3:122 SHANK3:146              |
| Renal salt wasting                       | -0.29337535 | 14        | 1.448e-04 | 1.768e-01 | KCNJ1:363 SCN1IA:442 CYP11A1:443 SARSZ:547 SCN11B:756 CYP21A2:1154 CLCNKB:1178 |
| Anemia, hereditary spherocytic hemolytic | 0.48520717  | 5         | 3.153e-04 | 2.093e-01 | ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA NA                         |
| Ciliopathies                             | 0.0862931   | 150       | 3.110e-04 | 2.093e-01 | TCTN1:34 TCTN2:182 CDC39:169 NPH3:375 NME8:275 ARL13B:282 TTN3:343             |
| Downturned corners of mouth              | -0.13956583 | 57        | 2.729e-04 | 2.093e-01 | HERC2:111 HDAC4:118 SNRNP:156 SMG1:166 CPLX1:346 SMCA1:383 KCNN1:428           |
| Hyponatremia                             | 0.18387534  | 32        | 3.214e-04 | 2.093e-01 | CLDN1:66 SCN11A:442 CYP11A1:443 COMT:449 SARSZ:547 MCCR7:38 SCN11B:756         |
| Liver carcinoma                          | -0.02556623 | 2215      | 2.144e-04 | 2.093e-01 | EP400:7 IGFBP3:13 SMPD3:23 GAL:40 TCF15:42 BNC2:50 DYNC1H1:53                  |
| Nephrocalcinosis                         | 0.15779439  | 45        | 2.537e-04 | 2.093e-01 | TRPM6:4 CYP2R1:21 CLDN1:66 SLC4A1:221 KCNN1:363 GRM2:295 SLC12A3:546           |
| Spherocytosis                            | 0.48520717  | 5         | 3.153e-04 | 2.093e-01 | ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA NA                         |
| Night blindness, congenital stationary   | -0.21906490 | 21        | 5.131e-04 | 2.785e-01 | PDE6B:74 USP11:91 GRP34:345 TRPM1:356 CABP4:454 GRM5:636 CACNA1F:831           |
| Autism Spectrum Disorders                | -0.05360059 | 369       | 4.825e-04 | 2.785e-01 | JAKMIP1:1 NBEA:                                                                |